

Lección 24

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Plan de trabajo

► Formulación hamiltoniana

— Teoría de perturbaciones

* Formulación sin dependencia temporal explícita
(variables eje-ángulo)

* Invariantes Adiabáticos

► Nuestro Syllabus:

FIS-310:

UNIDAD 5. Formulación Hamiltoniana

5.1. Transformaciones de Legendre

5.2. Ecuaciones de Hamilton

5.3. Ecuaciones de Hamilton de la Partícula Relativista

5.4. Corchete de Poisson

5.5. Transformaciones Canónicas

5.6. Ecuaciones de Hamilton-Jacobi

5.7. Teoría de perturbaciones

FIS-345:

Unidad IV: Formulación Hamiltoniana General

- Transformaciones canónicas
- Variables de ángulo y acción. Invariantes Adiabáticos

Unidad V: Teoría de perturbaciones y pequeñas oscilaciones

- Teoría canónica de perturbaciones
- Caos en sistemas Hamiltonianos

FIS-426:

4) Formulación Hamiltoniana General

- Paréntesis de Poisson
- Transformaciones canónicas
- Variables de ángulo y acción
- Teoría de perturbaciones
- Caos en sistemas Hamiltonianos

Canonical perturbation theory

Time-dependent vs time-independent CPT

–The Canonical Perturbation Theory (CPT) which we developed is called time-dependent:

*Find classical solution without perturbation, $\mathbf{p}_0(t; \alpha, \beta)$, $\mathbf{r}_0(t; \alpha, \beta)$

*Evaluate $\Delta H(t; \alpha, \beta) = \Delta H(\mathbf{p}_0(t; \alpha, \beta), \mathbf{r}_0(t; \alpha, \beta))$ and use it to solve equations

$$\dot{\alpha} = -\frac{\partial \Delta H}{\partial \beta}, \quad \dot{\beta} = \frac{\partial \Delta H}{\partial \alpha}$$

* Allows to get instant t -dependence of parameters $\alpha(t)$, $\beta(t)$

* Since $\alpha(t)$, $\beta(t)$ change weekly, in some problems we do not need detailed info about t -dependence, only need period-averaged quantities ...

** We may have only implicit equation for trajectory ...

** Evaluation of $\frac{1}{T} \int_0^T dt (...)$ may be not easy technically

–There is alternative formulation of CPT which allows to get certain effects (for example, change of period of motion, precession due to perturbation) directly, not using t -dependence explicitly

Last lecture: Action-angle variables

$$I_n = \oint \frac{p_n dq_n}{2\pi}, \quad w_n = \frac{\partial S_0}{\partial I_n}, \quad \underbrace{S_0(\mathbf{q}, I_1, \dots, I_n)}_{\text{Gener. function "F}_2(\mathbf{q}, \mathbf{P})"} = \sum_n \int^n p_n dq_n,$$

Canonical variables:

–variables I_n are called actions, w_a are called angles

$$\dot{I}_n = -\frac{\partial H}{\partial w_n} = 0, \quad \dot{w}_n = \frac{\partial H}{\partial I_n} = \text{const}, \quad \Rightarrow w_n = \left(\frac{\partial H}{\partial I_n} \right) t + \text{const}$$

$\Rightarrow$ **Hamiltonian depends only on $I_1, \dots, I_n$** , NOT on w .

–Variable I_n characterize shape of trajectory, w_n characterize orientation

–If we know $H(I_1, \dots, I_n)$, should be able to find all main characteristics of the system

*Curious observation: In semiclassical Bohr-Sommerfeld quantization rules,

$$I_n = k_n \hbar, \quad k_n \in \mathbb{N} \quad (\text{usually works for } k_n \gg 1)$$

–in action-angle variables, the **time dependence $w(t)$ is always linear**, and the action

$$S_0 = \sum_n I_n \dot{w}_n$$

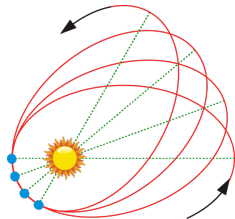
Some results in action-angle variables

Trajectory is always periodic,

$$\mathbf{q}(w_a + 2\pi) = \mathbf{q}(w_a)$$

in view of periodicity, coordinates q_a can be written as multidimensional Fourier series expansion

$$q_k = \sum_{\mathbf{j} \in \mathbb{Z}^n} a_{\mathbf{j}}^{(k)} e^{i\mathbf{j} \cdot \mathbf{w}}, \quad \mathbf{j} = \{j_1, \dots, j_n\}, \quad j_a \in \mathbb{Z}$$
$$a_{\mathbf{j}}^{(k)} = \int dw_1 \dots dw_N e^{-i\mathbf{j} \cdot \mathbf{w}} q_k$$



$q_a(t)$ is periodic if different w_a are commensurate (rational numbers of each other): $\forall a, b \exists j_a, j_b \in \mathbb{N}$,

$$j_a w_a = j_b w_b \quad (\text{no sum})$$

or, equivalently, the frequencies should be commensurate :

$$\left(\frac{\partial H}{\partial l_1} \right) : \left(\frac{\partial H}{\partial l_2} \right) : \left(\frac{\partial H}{\partial l_3} \right) = n_1 : n_2 : n_3, \quad n_{1,2,3} \in \mathbb{N}$$

In general the motion is “multi-periodic”,

$$\left(\frac{\partial H}{\partial l_1} \right) : \left(\frac{\partial H}{\partial l_2} \right) : \left(\frac{\partial H}{\partial l_3} \right) \neq n_1 : n_2 : n_3,$$

Examples

► Harmonic oscillator:

$$H_{\text{osc}} = E_1 + E_2 + E_3 = I_1\omega_1 + I_2\omega_2 + I_3\omega_3, \quad \omega_a = \sqrt{k_a/m}$$

–full degeneracy iff $\omega_1 = \omega_2 = \omega_3$, result becomes a function of $I_1 + I_2 + I_3$

$$\dot{w}_r : \dot{w}_\theta : \dot{w}_\phi = \frac{\partial H_{\text{osc}}}{\partial I_r} : \frac{\partial H_{\text{osc}}}{\partial I_\theta} : \frac{\partial H_{\text{osc}}}{\partial I_\phi} = 1 : 1 : 1$$

Trajectory : $q_a = \sqrt{2I_a/k_a\omega_a} \cos w_a$ (independent for each $a = x, y, z$)

► Coulomb potential:

$$H_{\text{Coulomb}} = -\frac{2\pi^2 mk^2}{(I_r + I_\theta + I_\phi)^2}$$

–full degeneracy for Coulomb potential, only $I_r + I_\theta + I_\phi$ contributes

$$\dot{w}_r : \dot{w}_\theta : \dot{w}_\phi = \frac{\partial H_{\text{Coulomb}}}{\partial I_r} : \frac{\partial H_{\text{Coulomb}}}{\partial I_\theta} : \frac{\partial H_{\text{Coulomb}}}{\partial I_\phi} = 1 : 1 : 1$$

any orbit is always closed

$$w_r = \frac{\psi - e \sin \psi}{2\pi}, \quad r = a(1 - e \cos \psi), \quad a = \frac{I_\theta + I_\phi}{4\pi^2 mk}, \quad e = \sqrt{1 - \left(\frac{I_\theta + I_\phi}{I_r + I_\theta + I_\phi} \right)^2}$$

Perturbation theory in action-angle variables

- Our current plan:

- Develop perturbation theory (very specific):

- * do not try to reconstruct trajectory $q(t)$, work only in terms of the action-angle variables (I_a, w_a) of the system

- * Assume hamiltonian $H = H_0 + \Delta H$, $\Delta H \ll H_0$

- * We can rewrite $H_0 = H_0(I_0)$ and need to find I -dependence of H , $H = H(I)$. We expect that all parameters of the orbit (shape etc) are encoded in the parameters I

- * Essentially in this approach drop the “coordinates” $t - t_0$, $\varphi - \varphi_0$... which determine orientation of the trajectory in space, time, ... ; focus on parameters which determine the orbit's size, shape, ...

- Since I , I_0 do not depend on time, such theory is called time-independent canonical perturbation theory

Perturbation theory in Action-angle variables

- Assume Hamiltonian $H = H_0 + \Delta H$, and we know $(I_0, \mathbf{w}_0)$ of H_0 , so

$$H_0 = H_0(I_0)$$

- we use vectorial notation $I = \{I_1, \dots, I_n\}$, $I_0 = \{I_1^{(0)}, \dots, I_n^{(0)}\}$ for brevity
- we need: reconstruct $H(I)$

► In terms of variables I , hamiltonian $H = H(I)$ may be expanded as

$$H = h(I, \epsilon) = h_0(I) + \epsilon h_1(I) + \dots, \quad \dot{\mathbf{w}} = \text{const}$$

- We may use $(I_a^{(0)}, w_a^{(0)})$ in a theory with $\Delta H \neq 0$:

$$H(I_0, \mathbf{w}_0) = H_0(I_0) + \epsilon H_1(I_0, \mathbf{w}_0) + \epsilon^2 H_2(I_0, \mathbf{w}_0) + \dots, \quad \epsilon \sim \Delta H/H \ll 1$$

- Now $H(I_0, \mathbf{w}_0)$ includes dependence on $\mathbf{w}_0$ since $(I_0, \mathbf{w}_0)$ are not his action/angle variables.

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For “angular frequencies” ω_a (we’ll use vector notation $\boldsymbol{\omega}$) we have

$$\boldsymbol{\omega} = \boldsymbol{\omega}_0 + \epsilon \frac{\partial h_1}{\partial I} + \epsilon^2 \frac{\partial h_2}{\partial I} + \dots$$

Perturbation theory in Action-angle variables

► Assume $Y = Y(\mathbf{w}_0, I)$ is a generating function of canonical transformation $(I_0, \mathbf{w}_0) \rightarrow (I, \mathbf{w})$

$$Y = \mathbf{w}_0 \cdot I + \epsilon Y_1(\mathbf{w}_0, I) + \epsilon^2 Y_2(\mathbf{w}_0, I) + \dots$$

$$I_0 = \frac{\partial Y}{\partial \mathbf{w}_0} = I + \epsilon \frac{\partial Y_1}{\partial \mathbf{w}_0} + \epsilon^2 \frac{\partial Y_2}{\partial \mathbf{w}_0} + \dots$$

$$\mathbf{w} = \frac{\partial Y}{\partial I} = \mathbf{w}_0 + \epsilon \frac{\partial Y_1}{\partial I} + \epsilon^2 \frac{\partial Y_2}{\partial I}$$

$$\Rightarrow H\left(\mathbf{w}_0, \frac{\partial Y}{\partial \mathbf{w}_0}\right) = h(I, \epsilon)$$

► Expand both sides over ϵ :

$$\Rightarrow H_0\left(\frac{\partial Y}{\partial \mathbf{w}_0}\right) + \epsilon H_1\left(\mathbf{w}_0, \frac{\partial Y}{\partial \mathbf{w}_0}\right) + \epsilon^2 H_2\left(\mathbf{w}_0, \frac{\partial Y}{\partial \mathbf{w}_0}\right) = h_0(I) + \epsilon h_1(I) + \epsilon^2 h_2(I) + \dots$$

Should take into account that Y also depends on ϵ and require that equation is satisfied order by order

Perturbation theory in Action-angle variables

$$H_0 \left(\frac{\partial Y}{\partial \mathbf{w}_0} \right) = H_0(I) + \left(\epsilon \frac{\partial Y_1}{\partial \mathbf{w}_0} + \epsilon^2 \frac{\partial Y_2}{\partial \mathbf{w}_0} \right) \frac{\partial H_0}{\partial I} + \frac{\epsilon^2}{2} \frac{\partial^2 H_0}{\partial I_a \partial I_b} \frac{\partial Y_1}{\partial w_{0a}} \frac{\partial Y_1}{\partial w_{0b}}$$

$$H_1 \left(\mathbf{w}_0, \frac{\partial Y}{\partial \mathbf{w}_0} \right) = H_1(\mathbf{w}_0, I) + \left(\epsilon \frac{\partial Y_1}{\partial \mathbf{w}_0} \right) \frac{\partial H_1}{\partial I}$$

$$\Rightarrow h_0(I) = H_0(I)$$

$$h_1(I) = H_1(\mathbf{w}_0, I) + \omega_0 \frac{\partial Y_1}{\partial \mathbf{w}_0} = \omega_0 \frac{\partial Y_1}{\partial \mathbf{w}_0} + \Phi_1(\mathbf{w}_0, I)$$

$$h_2(I) = H_2(\mathbf{w}_0, I) + \omega_0 \frac{\partial Y_2}{\partial \mathbf{w}_0} + \frac{\partial Y_1}{\partial \mathbf{w}_0} \frac{\partial H_1}{\partial I} + \frac{1}{2} \frac{\partial^2 H_0}{\partial I_a \partial I_b} \frac{\partial Y_1}{\partial w_{0a}} \frac{\partial Y_1}{\partial w_{0b}}$$
$$:= \Phi_2(\mathbf{w}_0, I) + \omega_0 \frac{\partial Y_2}{\partial \mathbf{w}_0}$$

If motion is periodic w.r.t. both $(I, \mathbf{w})$ and $(I_0, \mathbf{w}_0)$, then

$$Y_k \equiv \sum_{j \neq 0} B_j^{(k)}(I) \exp(i \mathbf{j} \cdot \mathbf{w}_0)$$

$\Rightarrow$ for quantities averaged over period $(\oint d\mathbf{w}_i)$ should keep only the terms with $\mathbf{j} = 0$:

$$h_0(I) = H_0(I), \quad h_1(I) = \overline{H_1(\mathbf{w}_0, I)}, \quad h_2(I) = \overline{\Phi_2(\mathbf{w}_0, I)}, \quad \dots$$

Example

Apply this method for evaluation of the frequency ω of the anharmonic oscillator

$$H = \frac{p^2}{2m} + \frac{m\omega_0^2 q^2}{2} \left(1 + \epsilon \frac{q}{q_0} \right)$$

Solution:

–Separate hamiltonian into H_0 (analytically solvable part) and perturbation:

$$H_0 = \frac{p^2}{2m} + \frac{m\omega_0^2 q^2}{2}, \quad H_1 = \frac{m\omega_0^2 q^3}{2q_0},$$

►Hamiltonian H_0 and trajectory in terms of its action-angles (I_0, w_0)

$$H_0 = \omega_0 I_0, \quad q_{(0)}(t) = \sqrt{\frac{2E}{q}} \sin \omega_0 t = \sqrt{\frac{I_0}{\pi m \omega_0}} \sin w_0$$

$$\Rightarrow H_1 = \frac{m\omega_0^2 q^3}{2q_0} = \frac{m\omega_0^2}{2q_0} \left(\frac{I_0}{\pi m \omega_0} \right)^{3/2} \sin^3 w_0$$

Example

We want to find $H = H(I) = h_0(I) + \epsilon h_1(I) + \epsilon^2 h_2(I)$

$$h_0(I) = H_0(I) = \omega_0 I_0, ,$$

$$h_1(I) = \overline{H_1(\mathbf{w}_0, I)} = (\dots) \int_0^{2\pi} \frac{dw_0}{2\pi} \sin^3 w_0 = 0 \Rightarrow h_1 = 0$$

$$h_2(I) = \overline{\Phi_2(\mathbf{w}_0, I)} = H_2(\mathbf{w}_0, I) + \int_0^{2\pi} \frac{dw_0}{2\pi} \frac{\partial Y_1}{\partial \mathbf{w}_0} \frac{\partial H_1}{\partial I} + \frac{1}{2} \frac{\partial^2 H_0}{\partial I_a \partial I_b} \frac{\partial Y_1}{\partial w_{0a}} \frac{\partial Y_1}{\partial w_{0b}}$$

where we took into account that H_0 is linear function and $H_2 = 0$.

Earlier we got

$$h_1(I) = H_1(\mathbf{w}_0, I) + \omega_0 \frac{\partial Y_1}{\partial \mathbf{w}_0}$$

so from $h_1 = 0$ conclude

$$\overline{\frac{\partial Y_1}{\partial \mathbf{w}_0}} = -\frac{1}{\omega_0} \overline{H_1(\mathbf{w}_0, I)}$$

Example

$$\Phi_2(w_0, I) = -\frac{1}{2\omega_0} \frac{\overline{\partial H_1^2}}{\partial I} = \dots = -\frac{3I^2}{4\pi^2 m q_0^2} \int_0^{2\pi} \frac{dw_0}{2\pi} \sin^6 w_0,$$

$$\langle \sin^6 w_0 \rangle = \frac{1}{2\pi} \int_0^{2\pi} dw_0 \sin^6 w_0 = \frac{5}{16}$$

$$h_2(I) = -\frac{15 I^2}{64\pi^2 m q_0^2}$$

$$\Rightarrow H(I) = \omega_0 I - \frac{15 \epsilon^2 I^2}{64\pi^2 m q_0^2}$$

$$\omega \equiv \dot{w} = \frac{\partial H}{\partial I} = \omega_0 - \frac{15 \epsilon^2 I}{16\pi^2 m q_0^2}$$

-the frequency oscillations changes. In the LO $I = E/\omega_0$, so may rewrite the correction as

$$\delta\omega = -\frac{15 \epsilon^2 E}{16\pi^2 m \omega_0 q_0^2}$$

(earlier we obtained this result using other methods).

Perturbation theory in Action-angle variables

Final result:

$$h_0(I) = H_0(I), \quad h_1(I) = \overline{H_1(w_0, I)}, \quad h_2(I) = \overline{\Phi_2(w_0, I)}, \quad \dots$$

Once we got $H(I) = h_0(I) + \epsilon h_1(I) + \dots$, can recover periods, precession, etc.

Example which we analyzed in the last lecture:

$$H = \frac{p^2}{2m} + \frac{m\omega_0^2 q^2}{2} \left(1 + \epsilon \frac{q}{q_0} \right)$$

Algorithm:

- 1 Find the trajectory $q_0(t, \alpha)$ in Hamiltonian H_0
- 2 Express constants α in terms of invariants I , and time dependence t with w
- 3 Put this into H_1 , find the average $\overline{H_1} = \int_0^{2\pi} dw H_1(w, I)$
- 4 Use (1) to find $h_0, h_1, \dots$
- 5 Take derivative $\partial h_1 / \partial I, \partial h_2 / \partial I$ to find corrections $\delta\omega$ to frequency ...

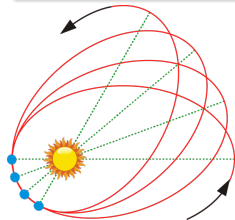
(looks excessively complicated for such simple example)

Example 2: Corrections to Kepler problem

Assume that a particle moves in the central field

$$U(r) = -\frac{\alpha}{r} + \varepsilon \delta V(r), \quad \alpha > 0$$

Develop a perturbation theory assuming that $\varepsilon \ll 1$, and find precession of the orbit due to $\mathcal{O}(\varepsilon)$ -correction



Corrections to Kepler problem

We found earlier that in Kepler problem ($\varepsilon = 0$)

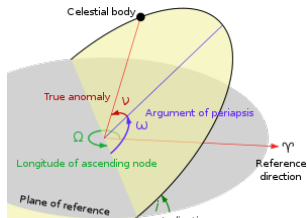
$$H_0(I_0) = -\frac{2\pi^2 mk^2}{(I_{r0} + I_{\theta 0} + I_{\phi 0})^2}$$

and trajectory is given by

$$w_r = \frac{\psi - e \sin \psi}{2\pi}, \quad r = a(1 - e \cos \psi), \quad a = \frac{I_{\theta} + I_{\phi}}{4\pi^2 mk},$$

- Using trajectory, may find $H_1(r) = H_1(I_{r0}, \mathbf{w}_0)$
- May extract dependence on I using expressions for quantities averaged over period ($\oint dw_i$)

$$h_0(I) = H_0(I), \quad h_1(I) = \overline{H_1(\mathbf{w}_0, I)} = \int_0^{2\pi} \prod_a \frac{dw_a}{2\pi} V(r(I, \mathbf{w}))$$



- For integration we may use Delaunay's angles
 - $\psi(t)$ is the polar angle of particle *in plane of motion*
 - i is the *inclination* - angle between plane of motion and plane $\perp$ to axis of spheroid
 - Ω is the longitude of ascending node
 - We need to rewrite trajectory as a function of $w_r, w_{\theta}, w_{\phi}, \dots$ take integrals

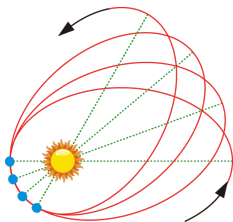
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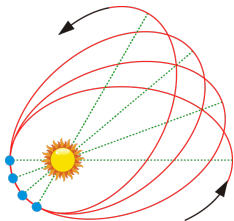
$$w_r = \frac{\psi - e \sin \psi}{2\pi}, \quad r = a(1 - e \cos \psi), \quad a = \frac{I_\theta + I_\phi}{4\pi^2 m k},$$



–Using trajectory, may find

$$h_0(I) = H_0(I)$$

$$h_1(I) = \overline{H_1(\mathbf{w}_0, I)} = \int_0^{2\pi} \prod_a \frac{dw_a}{2\pi} V(r)$$



–only dependence on I_r will remain; result is no longer function of only $I_r + I_\theta + I_\phi$

–Perturbation eliminated degeneration which we had

► Important consequence:

$$\dot{w}_r : \dot{w}_\phi = \frac{\partial H}{\partial I_r} : \frac{\partial H}{\partial I_\phi} \neq 1 : 1$$

Adiabatic invariants

Adiabatic invariants

Previous studies:

Perturbation theory in presence of explicit perturbation $\Delta H(t)$

This lecture:

- we do NOT have explicit perturbation
- We know that one of the parameters λ of the system *slowly* depends on time in a known way, $\lambda = \lambda(t)$
- We know exact solution for $\lambda = \lambda_0 = \text{const}$, yet we do not know solution for $\lambda(t)$
- Formally, we *could* consider $\Delta H = H(p, q, \lambda(t)) - H(p, q, \lambda_0)$ as perturbation, and repeat what we did earlier. However, it is not the optimal way, since it would be valid only for $\Delta H \ll H$ (and slow change of $\lambda(t)$ does not imply that total change $\Delta\lambda \ll \lambda$)

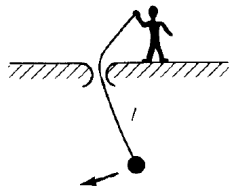
Adiabatic invariants

Assume the hamiltonian of the problem H depends on some parameter λ , and we know solution when $\lambda = \text{const}$

– Assume the parameter λ now changes slowly as a function of time, so that $T \dot{\lambda} \ll \lambda$, where T is the period of oscillation/characteristic scale.

Example

Oscillations of mathematical pendulum, length slowly changes as a function of time, $\ell = \ell(t)$. How does the energy change as a function of time ?



Adiabatic invariants

If $\lambda = \text{const}$, energy is conserved, so for $\lambda = \lambda(t)$ change of energy:

$$\frac{dE}{dt} = \frac{\partial H}{\partial t} = \frac{\partial H}{\partial \lambda} \dot{\lambda}$$

Let's average over the period (assume $\dot{\lambda}(t)$ is almost a constant):

$$\overline{\frac{dE}{dt}} = \frac{\dot{\lambda}}{T} \int_0^T dt \frac{\partial H}{\partial \lambda}$$

Hamiltonian equations: $\dot{q} \equiv dq/dt = \partial H / \partial p$, so

$$\dot{q} = \frac{\partial H}{\partial p}, \quad \Rightarrow \quad dt = \frac{dq}{\partial H / \partial p}$$

$$T = \int_0^T dt = \oint dq \frac{1}{\partial H / \partial p} \quad (\text{period})$$

$$\overline{\frac{dE}{dt}} = \frac{\dot{\lambda}}{T} \int_0^T dt \frac{\partial H}{\partial \lambda} = \dot{\lambda} \frac{\oint dq \frac{\partial H / \partial \lambda}{\partial H / \partial p}}{\oint dq \frac{1}{\partial H / \partial p}}$$

We integrate over the whole trajectory $E = \text{const} (\lambda = \text{const})$

$$\Rightarrow p = p(q, E, \lambda)$$

along the trajectory.

Adiabatic invariants

Since $E \approx \text{const}$,

$$\Rightarrow 0 = \frac{dH}{d\lambda} = \frac{\partial H}{\partial \lambda} + \frac{\partial H}{\partial p} \frac{\partial p}{\partial \lambda}$$
$$\Rightarrow \frac{\partial H / \partial \lambda}{\partial H / \partial p} = -\frac{\partial p}{\partial \lambda}, \quad \frac{1}{\partial H / \partial p} = \frac{\partial p}{\partial E}$$

similarly,

$$\frac{dE}{dt} = \dot{\lambda} \frac{\oint dq \frac{\partial H / \partial \lambda}{\partial H / \partial p}}{\oint dq \frac{1}{\partial H / \partial p}} = -\dot{\lambda} \frac{\oint dq \frac{\partial p}{\partial \lambda}}{\oint dq \frac{\partial p}{\partial E}}$$
$$\Rightarrow \oint dq \left(\frac{\partial p}{\partial \lambda} \dot{\lambda} + \frac{\partial p}{\partial E} \frac{dE}{dt} \right) = 0, \quad \Rightarrow \quad \frac{d}{dt} I = 0, \quad I(E, \lambda) = \oint \frac{p dq}{2\pi}$$

where I is the action variable for problem with $\lambda = \text{const}$. We remember that

$$\frac{\partial I}{\partial E} = \frac{1}{2\pi} \oint \frac{\partial p}{\partial E} dq = \frac{1}{2\pi} \int_0^T dt = \frac{T}{2\pi} = \frac{1}{\omega}$$

in agreement with our earlier findings, $\dot{\omega} \equiv \omega = \partial E / \partial I$

—Earlier we've shown that $I = E / \omega$

$\Rightarrow$ in case of slow change of λ , $I = E(t) / \omega(t)$ doesn't change, i.e.

$$E(t) \sim \omega(t) = \sqrt{g / \ell(t)}$$

Adiabatic invariants

- ▶ We found condition $\bar{I} = \text{const}$ using very simplistic derivation
 - We liberally permuted averaging over period $(\dots)$ and d/dt at several steps
- ▶ Repeat the derivation using more formal description in terms of action-angle variables (which allows generalization for N degrees of freedom)
 - Assume (I_0, w_0) are action-angles variables for $\lambda = \text{const}$, and (I, w) for time-dependent $\lambda(t)$.
 - Assume $W(I, w_0, \lambda)$ is a generating function of the canonical transform $(p, q) \rightarrow (I, w)$
 - since $\partial_t W \neq 0$, the hamiltonian changes as

$$H \rightarrow H(I_0, w_0) + \frac{\partial W}{\partial t} = H(I_0, w_0) + \dot{\lambda} \frac{\partial W}{\partial \lambda} = H(I_0, w_0) + \Delta H(t)$$

- We'll treat the last term as perturbative time-dependent correction ΔH which affects evolution of I_0 :

$$\dot{I}_0 = -\frac{\partial \Delta H}{\partial w_0} = -\dot{\lambda} \frac{\partial}{\partial w_0} \left(\frac{\partial W}{\partial \lambda} \right) \neq 0 \quad (?!)$$

Adiabatic invariants

► Average over the period:

$$\overline{\frac{dl_0}{dt}} = -\frac{1}{T} \int_0^T dt \dot{\lambda} \frac{\partial}{\partial w_0} \left(\frac{\partial W}{\partial \lambda} \right) \approx -\frac{\dot{\lambda}}{T} \int_0^T dt \frac{\partial}{\partial w_0} \left(\frac{\partial W}{\partial \lambda} \right)$$

For periodic motion

$$\frac{\partial W}{\partial \lambda} = \sum_j A_j(l_0) e^{ij \cdot w_0}$$

- The term with $j = 0$ does not depend on w_0 , so it does not contribute to $\frac{\partial}{\partial w_0} \left(\frac{\partial W}{\partial \lambda} \right)$
- The contribution of the terms with $j \neq 0$ ($j \in \mathbb{N}, j = 1, 2, \dots$) vanishes after averaging over a period $\int_0^T dt \rightarrow \int_0^{2\pi} dw_0$

$$\Rightarrow \overline{\frac{dl_0}{dt}} = 0 \quad (!!!) \quad \text{as expected}$$

Control question

- What is the difference between the first integral and the adiabatic invariant?
- Do these concepts coincide?

Control question 2

- A system consist of a massive (infinitely heavy) star and N planets which rotate around it. The interaction of all planets with each other is very weak. All planets move in the plane.
- Identify how many first integrals and adiabatic invariants has this system

Example: Harmonic oscillator

Harmonic oscillator:

$$H = \frac{p^2}{2m} + \frac{m\omega^2(t)q^2}{2}$$

where $\omega = \omega(t)$ is slowly changing [we'll treat it as parameter $\lambda(t)$].

–For $\omega = \text{const}$, the generating function of $(p, q) \rightarrow (I_0, w_0)$ transformation was introduced in Lecture 15:

$$W(q, w_0) = \frac{m\omega q^2}{2} \cot w_0, \quad \frac{\partial W}{\partial \omega} = \frac{mq^2}{2} \cot w_0$$

$$p = m\omega q \cot w_0 = \frac{\partial W}{\partial q}, \quad I_0 = \frac{1}{2\pi} \oint p dq = \frac{\pi m\omega}{\sin^2 w_0} = -\frac{\partial W}{\partial w_0}$$

$$\frac{\partial W}{\partial \omega} = \frac{I_0}{4\pi\omega} \sin 2w_0$$

$$\Rightarrow \overline{\frac{dI_0}{dt}} = \frac{\dot{\omega}}{T} \int_0^T dt \frac{\partial}{\partial w_0} \left(\frac{\partial W}{\partial \omega} \right) = -\frac{\dot{\omega}}{\omega} \overline{I_0 \cos 2w_0} = 0 \quad \text{as expected}$$

Example 2

A planet of mass m orbits around a star of mass $M \gg m$. Due to radiation, the mass of the star slowly decreases, $\dot{M}(t) < 0$, $|\dot{M}| \ll M/T$, where T is the period of orbital motion of the planet. Analyze how the trajectory of the planet will change due to decrease of the star's mass. Formulate your reply in terms of the orbit's conventional parameters: major semi-axis, eccentricity, period of rotation and position of periapsis (="closest point").

Help: Decrease of the star's mass might be reinterpreted as slow weakening of the attractive gravitational potential

$$U_{12} = -\frac{G M(t) m}{r_{12}} = -\frac{k(t)}{r}, \quad |\dot{k}| \ll k/T$$

Before we start ...

$$U_{12} = -\frac{G M(t) m}{r_{12}} = -\frac{k(t)}{r}, \quad \left| \dot{k} \right| \ll k/T$$

Analyze expected trajectory in this potential (without assumption $\left| \dot{k} \right| \ll k/T$):

- *Are there any continuous symmetries in this potential?*
 - ▶ *Are there any integrals of motion related to this symmetries?*
- *What can we say about trajectories? Do Kepler's laws remain valid?*



Example 2

Solution:

- Motion occurs in plane: no forces in direction transverse to plane of motion, central problem $U(r, t)$ for any t -dependence
 - Effective 2D motion in a plane
 - Choose polar coordinates (r, θ) .
 - Since $M \gg m$, center of mass of the system coincides with that of the star
 - For problems of this type the best method is to use the conservation of adiabatic invariants

$$I_r \equiv \oint \frac{p_r dr}{2\pi}, \quad I_\theta = \oint \frac{p_\theta d\theta}{2\pi}$$

in case of slow (adiabatic) change of the parameters of the system, so the idea of the solution is to express the parameters of the orbit in terms of these adiabatic invariants.

Example 2

Last lecture: we found the expressions in Coulomb field

$l_\theta = L = \text{const} = \text{angular momentum}$

$$l_r = -l_\theta + \pi k(t) \sqrt{\frac{2m}{|E|}} = \text{const} = \alpha$$

$\Rightarrow$ angular momentum will conserve, but energy won't since the system is not conservative (potential depends on time),

$$\Rightarrow E(t) = -\frac{2\pi^2 m^2 k^2(t)}{(l_r + l_\theta)^2} \sim M^2(t)$$

- The absolute value of energy will decrease as a square of mass (!).
- Similarly can analyze other parameters of the orbit.

Example 2

– Similarly can analyze other parameters of the orbit. We know that for $M = \text{const}$ the trajectory is given by

$$\frac{p}{r(\phi)} = 1 + e \cos(\phi - \phi_0)$$

where e is eccentricity and p is dimensional parameter related to major semi-axis a and minor semi axis b as

$$a = \frac{r(\phi = \phi_0) + r(\phi = \phi_0 + \pi)}{2} = \frac{p}{1 - e^2}$$

$$b = a\sqrt{1 - e^2} = \frac{p}{\sqrt{1 - e^2}}$$

The parameters p , e are directly related to energy and angular momentum as

$$p = \frac{L^2}{mk(t)}, \quad e = \sqrt{1 + \frac{2 \cancel{E(t)} L^2}{m \cancel{k^2(t)}}} = \sqrt{1 - \frac{4\pi^2 m L^2}{(l_r + l_\theta)^2}}$$

$\Rightarrow$ When $M(t)$ decreases, the eccentricity e will not change, but the parameter p and major semi-axis will increase as $\sim 1/k(t) \sim 1/M(t)$.

Example 2

The period of rotation might be found from the 3rd law of Kepler:

$$\begin{aligned} T &= \oint dt = \oint \frac{d\phi}{\dot{\phi}} = \frac{m}{L} \oint \underbrace{r^2 d\phi}_{\text{area}} = \frac{\pi ab m}{L} = \\ &= \frac{\pi m p^2(t)}{L(1-e^2)^{3/2}} = \frac{\pi L}{m(1-e^2)^{3/2} k^2(t)} \sim \frac{1}{M^2(t)} \end{aligned}$$

-gradually *increases*:

—in weakening field, the planet moves further and further from the start, so angular momentum conservation requires that that planet moves slower.

Example 2

– The rotation of periaapsis occurs due to time-dependence of the mass $M(t)$; if $M = \text{const}$ then there is no rotation. Let's introduce

$$\Delta H = H(t) - H(t=0) = -\frac{G (M(0) - M(t)) m}{r_{12}} = -\frac{(k(0) - k(t))}{r_{12}}.$$

Since the time dependence is very slow, $(M(0) - M(T)) \ll M$, so we can treat it as a perturbation. Applying (time-dependent) canonical perturbation theory, we may find for the period-averaged precession of periaapsis

$$\overline{\dot{\theta}_0} = \frac{\partial \overline{\Delta H}}{\partial L} \sim \frac{\partial}{\partial L} \left(\int_0^T dt \frac{(k(0) - k(t))}{r(t)} \right) \approx \dot{k}(0) \frac{\partial}{\partial L} \left(\int_0^T dt \frac{t}{r(t)} \right)$$

$$(k(0) - k(t)) \approx -t \dot{k}(0)$$

where we used Taylor expansion in numerator, $k(0) - k(t) \approx -t \dot{k}(0)$. Since we know explicitly the dependence $r(t)$ in the Coulomb field, we may evaluate this integral

$$\begin{aligned} \left(\int_0^T dt \frac{t}{r(t)} \right) &= \int_0^{2\pi} d\xi a \sqrt{\frac{ma}{k(t)}} (1 - e \cos \xi) \frac{a \sqrt{\frac{ma}{k(t)}} (\xi - e \sin \xi)}{a(1 - e \cos \xi)} = \\ &= \frac{ma^2}{k(t)} \int_0^{2\pi} d\xi (\xi - e \sin \xi) = \frac{2\pi^2 ma^2}{k(t)} = \frac{L^2 (I_r + I_\theta)^2}{2m^2 k^3(t)} \neq 0 \end{aligned}$$

Example 3

Assume that a particle moves in homogeneous magnetic field $\mathbf{B}$. In case $\mathbf{B} = \text{const}$ the particle rotates around magnetic field: moves over a circumference or helicoidal line with constant radius. Analyze the motion of particle in case when the field $\mathbf{B}$ decreases slowly (adiabatically) as a function of time. The direction of the field does not change, only its magnitude decreases.

Help: Lagrangian in magnetic field is given by

$$L = -mc^2 \sqrt{1 - \frac{v^2}{c^2}} - q(\phi - \mathbf{A} \cdot \mathbf{v}) \approx -mc^2 + \frac{mv^2}{2} + q\mathbf{A} \cdot \mathbf{v}, \quad \mathbf{A} = \frac{1}{2} [\mathbf{r} \times \mathbf{B}]$$

Before we start ...

$$L = -mc^2 \sqrt{1 - \frac{v^2}{c^2}} - q(\phi - \mathbf{A} \cdot \mathbf{v}) \approx -mc^2 + \frac{mv^2}{2} + q\mathbf{A} \cdot \mathbf{v}, \quad \mathbf{A} = \frac{1}{2} [\mathbf{r} \times \mathbf{B}]$$

$$\mathbf{B} = B(t)\hat{\mathbf{z}}, \quad \mathbf{F} = q\mathbf{v} \times \mathbf{B}$$

Analyze expected trajectory in this field (without assumption $|\dot{\mathbf{B}}| \ll |\mathbf{B}|/T$):

- Are there any continuous symmetries in this system?
 - Are there any integrals of motion related to this symmetries?
- What can we say about trajectories?

Time left:

60

Example 3

► Hamiltonian in constant magnetic field:

$$p_a = \frac{\partial L}{\partial \dot{x}_a} = m \cdot \dot{x}_a + qA_a$$

$$\begin{aligned} H = p_a \cdot \dot{x}_a - L &= (m \cdot \dot{x}_a + qA_a) \dot{x}_a - \left(\frac{m \cdot \dot{x}_a^2}{2} + qA_a \dot{x}_a \right) \\ &= \frac{m\dot{x}^2}{2} = \frac{(\mathbf{p} - q\mathbf{A})^2}{2m} \end{aligned}$$

► Assume $\hat{z}$ points in direction of $\mathbf{B}$:

$$L = \frac{m(\dot{x}^2 + \dot{y}^2 + \dot{z}^2)}{2} + \frac{qB}{2}(\dot{x}y - \dot{y}x)$$

$$H = \frac{(\mathbf{p} - q\mathbf{A})^2}{2m} = \frac{(p_x - \frac{q}{2}yB)^2 + (p_y + \frac{q}{2}xB)^2 + p_z^2}{2m}$$

Example 3

► Equations of motion for $B = \text{const}$:

$$m\ddot{x} = -\frac{qB}{2}\dot{y}, \quad m\ddot{y} = \frac{qB}{2}\dot{x}, \quad m\ddot{z} = 0$$

Evaluate (1a) + i (1b), use variable $\xi = x + iy$:

$$\ddot{\xi} = i\frac{qB}{2m}\dot{\xi}, \quad x = \text{Re}(\xi), \quad y = \text{Im}(\xi)$$

$$\Rightarrow \xi = \xi_0 \exp\left(i\frac{qB}{2m}t\right) + \xi_1$$

-oscillations with cyclotron frequency around point $\xi_1 = x_1 + iy_1$

► Generalized momenta:

$$p_x = m\dot{x} + qBy/2, \quad p_y = m\dot{y} + qBx/2$$

$$\Rightarrow p_x + ip_y = m\dot{\xi} + \frac{qBi}{2}\xi = iqB\xi$$

$$\oint p_x dq_x = \oint p_y dq_y = \pi qB R_{\text{max}}^2$$

Example 3

$$\oint p_x dq_x = \oint p_y dq_y = \pi q B R_{\max}^2$$

When $B = B(t)$ decreases as a function of time, expect $R_{\max}$ should grow as $\sim 1/\sqrt{B(t)}$ to guarantee that $I_x = I_y \approx \text{const}$

► Recall that radius $R_{\max}$ is related to (transverse to $\mathbf{B}$) velocity $|v|$ as

$$|v| = \frac{qB}{2m} R_{\max},$$
$$\Rightarrow \frac{m |v_{\perp}|^2}{2} = \frac{q^2 B^2 R_{\max}^2}{8m} = \frac{qB(t) I_x}{8m}$$

conservation of I_x implies that mechanical energy of charge is not conserved (why ???)

However, the “magnetic moment” $\mu \equiv mv_{\perp}^2/2B \sim I_x$ is adiabatic invariant

Example 3

► Analysis in cylindrical coordinates ($\mathbf{B} = B\hat{\mathbf{z}} = \text{const}$):

$$L = \frac{m}{2} (\dot{r}^2 + r^2 \dot{\phi}^2 + \dot{z}^2) - \frac{q B r^2 \dot{\phi}}{2}$$

Angular momentum of a particle:

$$M_z = [\mathbf{r} \times \mathbf{p}]_z = m r^2 \dot{\phi}$$

► Variables z , ϕ are cyclical, two integrals of motion:

$$p_z = m\dot{z}, \quad p_\phi \equiv \frac{\partial L}{\partial \dot{\phi}} = m r^2 \left(\dot{\phi} - \frac{q B}{2m} \right)$$

For a particle which moves over circumference of radius R with center at origin, the angular frequency $\dot{\phi} = \omega_c$, so can get

$$p_\phi = m R^2 \left(\omega_c - \frac{q B}{2m} \right) = R^2 \frac{q B}{2}$$

► Motion over ϕ is periodic, so adiabatic invariant $I_\phi \equiv \oint p_\phi d\phi = \pi R^2 q B$

– May be related to “magnetic moment” $\mu \equiv m v_\perp^2 / 2B$

Control question

We discussed that:

- $v_z = \text{const}$ due to equations of motion
- Conservation of $\mu = mv_{\perp}^2/2B$ implies that $|v_{\perp}(t)| \sim \sqrt{B_z(t)}$ is not conserved.

Apparently this implies that energy of the particle is not conserved. However, we discussed that in magnetic field the force $\mathbf{F} = q\mathbf{v} \times \mathbf{B}$ is always orthogonal to $\mathbf{v}$ and thus should not change energy (!).

- *What causes change of the particle's energy which we found ???*



Example 3

- Analysis for slowly changing field ($\mathbf{B} = B(t)\hat{\mathbf{z}} = \text{const}$):
- Motion over ϕ is periodic, so $I_\phi \equiv \oint p_\phi d\phi = \pi R^2 q B$
- Now consider general case of slowly (adiabatically) decreasing $B(t)$:
—adiabatic invariant:

$\Rightarrow$

$I_\phi = \pi q R^2(t) B(t) = \pi q m R^2(t) \omega_c^2(t) (B(t)/m \omega_c^2(t)) = m v_\perp^2 / B(t) = \text{const}$
should *increase* as $1/\sqrt{B(t)}$, in agreement with earlier result in Cartesian coordinates.

Example 3 - extension

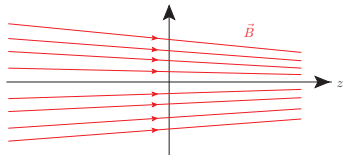
Assume that a particle moves in time-independent slightly inhomogeneous magnetic field $\vec{B}$ shown in the Figure ($B_x, B_y \ll B_z$, and B_z depends mildly on z). For momentum components of particle at time $t = 0$ we may assume $p_z \gg p_x, p_y$.

► This is not adiabatic problem “per se”, but can be reduced to such if we choose frame moving in direction z with velocity $V_z = p_z/m$

— In this frame we can consider that $B(t)$ slightly depends on time and apply what we learned earlier:

$$\mu \equiv \frac{m v_{\perp}^2}{2 B_z(t)} \sim I_{\phi} \approx \text{const}$$

so the particle will move over helicoidal line, but radius $R(t)$ will slowly increase as $\sim 1/\sqrt{B_z(t)}$ and transverse velocity will decrease as $\sqrt{B_z(t)}$



Adiabatic invariants vs parametric resonance

We found that small changes of $\lambda(t)$ do NOT change adiabatic invariants

$$I_a(t) = \oint \frac{p_a dq_a}{2\pi},$$

► Average over the period:

$$\overline{\frac{dI_a}{dt}} = 0$$

Beware

The conservation of adiabatic invariants will not work for systems close to *parametric resonance*, when small $\lambda(t)$ might cause huge oscillations

Questions & Answers (Q&A)

Please let me know if You have any questions or comments ...